

Contradiction Fields: CRT Rigidity, Prime Matrix Geometry, and Zeta-Zero Escape

Project manuscript

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Abstract

This monograph draft unifies two contradiction-field research programs. The first studies prime existence in CRT prime matrices through row, column, diagonal, anchor, and rough-composite rigidity. The second studies an off-critical zero of the Riemann zeta function through a global contradiction-field ledger for prime anomalies. The common method is to convert a hypothetical counterexample into a structured covering field, decompose it into local rigidity layers, and exclude all free escape channels by capacity, periodicity, and no-cycle bookkeeping.

This is a synthesis and dependency manuscript. It does not assert that the Riemann Hypothesis or the prime-matrix row/column theorem has already been accepted as a final unconditional theorem. Each claim is marked by status in the accompanying status table: proved in manuscript, reduced to a structured input, external, computationally certified, or requiring independent referee verification.

Contents

1	Guiding Principle: The Contradiction Field	7
2	Prime-Matrix Contradiction Fields	9
3	Zeta-Zero Contradiction Fields	11
4	Unified Dependency Map	13
5	Current Status and Honest Boundary	15

Chapter 1

Guiding Principle: The Contradiction Field

A contradiction field is a structured ledger attached to a hypothetical counterexample. Instead of estimating a single exceptional set, it records every possible explanation for the counterexample as a named channel. A proof closes when every channel is either absorbed, forced to descend in a finite potential, transferred with source deletion, or contradicts a global capacity lower/upper bound.

The common rigidities used throughout this manuscript are:

1. CRT skeletons and nonzero-residue balance;
2. local short-window non-reuse of large prime factors;
3. small-prime shell migration and diagonal locking;
4. rough-composite and double-anchor decompositions;
5. capacity ledgers after baseline subtraction;
6. no-cycle potentials for internal escape mechanisms.

Chapter 2

Prime-Matrix Contradiction Fields

Let P be an odd prime and write the $P \times P$ prime matrix entries as

$$n(r, c) = (r - 1)P + c, \quad 1 \leq r, c \leq P.$$

The row and column programs attempt to show that every nontrivial row or column contains a prime. The current formal appendix reduces a row or non- P column counterexample to a structured bad configuration after removing small-factor locks and tail anchors.

Theorem 2.1 (A/B reduction, recorded form). *A nontrivial full-composite row or column, after small-factor and tail-anchor peeling, induces a Structured-EHPD bad configuration satisfying CRT balance, large-factor non-reuse, and anchor-capacity constraints.*

Reference. See `docs/row-column-reduction-formal-appendix.md`. This theorem is a reduction statement; it does not by itself prove the Structured-EHPD exclusion theorem. $\square$

Chapter 3

Zeta-Zero Contradiction Fields

Assume an off-critical zero $\rho = \beta + i\gamma$ with $\beta > 1/2$. The RH contradiction-field manuscript transfers the resulting smooth prime anomaly into a CRT rough-composite ledger, decomposes it into sparse/dense/tail/internal/global exits, and proves that no named escape channel remains under the recorded local theorems and restricted external inputs.

Theorem 3.1 (RH contradiction-field synthesis, recorded form). *Under the numbered local theorems, controlled-exit lemmas, threshold hierarchy, and restricted external inputs recorded in `paper/rh-proof/rh-contradiction-field.tex`, an off-critical zero has no free escape channel in the contradiction-field event graph.*

Reference. See the RH manuscript in `paper/rh-proof/` and the final audit in `docs/`. The submission warning in the RH manuscript remains in force until independent line-by-line referee verification and final monograph page-number checks are completed. $\square$

Chapter 4

Unified Dependency Map

The two programs share the same logic:

counterexample $\Rightarrow$ structured covering field $\Rightarrow$ named exits $\Rightarrow$ capacity/no-cycle contradiction.

The prime-matrix program is more geometric and finite-CRT in nature; the zeta-zero program is analytic and scale-asymptotic. The monograph treats both as projections of the same contradiction-field method.

Layer	Prime-matrix projection	RH projection
Baseline	CRT nonzero class balance	smoothed CRT candidate baseline
Local rigidity	large-factor non-reuse, diagonal locks	sparse/dense/tail routing
Capacity	anchor coverage capacity	projection and square-function capacity
No-cycle	recursive peeling and shell migration	GEE transfer/no-cycle ledger
External inputs	finite verification and Structured-EHPD	explicit formula, KL, Vaaler, BG/Baker, sieve, Vaughan
Final status	reduction package, not final unconditional theorem	verification package with submission warning

Chapter 5

Current Status and Honest Boundary

The purpose of this monograph draft is to make the shared structure visible and auditable. It is not yet a final unconditional theorem manuscript. The precise statuses are maintained in `docs/monograph/claim-status-table.md`. Any future promotion to final theorem status requires independent verification of every reduced structured input and every controlled exit.